

RQ3 and RQ4: How are women and men in AI described?

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Contents

1: Load the data

Columns are renamed so the code is easier to read, and only people identified as female or male are kept.

```
people <- read_csv("data/people.csv") |>
  rename(wikidata_id = qid, name = title, intro = intro_text) |>
  filter(gender %in% c("female", "male")) |>
  mutate(era = factor(era, levels = c("pre-1950", "1950-69", "1970-89", "1990+")),
         gender = factor(gender, levels = c("female", "male")))

glimpse(people)
```

```
## Rows: 917
## Columns: 16
## $ wikidata_id    <chr> "Q3022082", "Q3022141", "Q92665", "Q92686", "Q92691", ~
## $ name           <chr> "Demetri Terzopoulos", "Demis Hassabis", "John F. Sowa~
## $ gender         <fct> male, male, male, male, male, male, male, male, male, ~
## $ description    <chr> "Distinguished Professor of Computer Science at UCLA",~
## $ birth_year     <dbl> 2000, 1976, 1940, 2000, 1935, 1966, 1956, 1928, 1947, ~
## $ sitelinks       <dbl> 8, 59, 8, 5, 9, 4, 5, 7, 3, 11, 3, 4, 2, 4, 3, 4, 2, 3~
## $ era            <fct> 1990+, 1970-89, pre-1950, 1990+, pre-1950, 1950-69, 19~
## $ gender_group   <chr> "male", "male", "male", "male", "male", "male", "male"~
## $ intro          <chr> "Demetri Terzopoulos is a Greek-Canadian-American com~
## $ article_bytes   <dbl> 19378, 90964, 11422, 4589, 3401, 3140, 4160, 7381, 581~
## $ created        <dtm> 2014-09-16 06:49:02, 2005-11-28 00:24:44, 2002-10-11 ~
## $ created_year   <dbl> 2014, 2005, 2002, 2004, 2006, 2008, 2007, 2009, 2017, ~
## $ recognition_lag <dbl> 14, 29, 62, 4, 71, 42, 51, 81, 70, 77, 87, 57, 40, 51,~
## $ pageviews_12m  <dbl> 2792, 1100406, 3154, 999, 1743, 2738, 522, 10229, 864,~
## $ in_degree      <dbl> 1, 44, 0, 1, 11, 5, 2, 1, 1, 0, 0, 2, 1, 0, 0, 1, 0, 9~
## $ out_degree     <dbl> 1, 38, 0, 2, 1, 4, 1, 1, 1, 5, 0, 1, 0, 0, 1, 1, 0, 14~
```

```
nrow(people)
```

```
## [1] 917
```

Group sizes: number of women and men in each birth era.

```
people |>
  count(era, gender) |>
  pivot_wider(names_from = gender, values_from = n, values_fill = 0)
```

```
## # A tibble: 5 x 3
##   era      female  male
##   <fct>    <int> <int>
## 1 pre-1950     23   154
## 2 1950-69      60   264
## 3 1970-89      56   135
## 4 1990+         8    37
## 5 <NA>        71   109
```

2. Clean the text

a. Remove unusable text

```
n_missing <- sum(is.na(people$intro) | str_trim(people$intro) == "")
n_missing
```

```
## [1] 7
```

```
people_text <- people |>
  filter(!is.na(intro), str_trim(intro) != "")
```

b. Words of each person's own name, so they can be removed from their own intro.

```
name_tokens <- people_text |>
  select(wikidata_id, name) |>
  unnest_tokens(word, name) |>
  distinct()

custom_stop <- tibble(word = c("born", "also", "known"))
```

c. Tokenise and clean

```
tokens <- people_text |>
  select(wikidata_id, gender, era, intro) |>
  unnest_tokens(word, intro) |>
  filter(!str_detect(word, "[0-9]"), str_length(word) > 2) |>
  anti_join(stop_words, by = "word") |>
  anti_join(custom_stop, by = "word") |>
  anti_join(name_tokens, by = c("wikidata_id", "word"))

head(tokens)
```

```
## # A tibble: 6 x 4
##   wikidata_id gender era   word
##   <chr>      <fct> <fct> <chr>
## 1 Q3022082   male   1990+ greek
## 2 Q3022082   male   1990+ canadian
## 3 Q3022082   male   1990+ american
## 4 Q3022082   male   1990+ computer
## 5 Q3022082   male   1990+ scientist
## 6 Q3022082   male   1990+ entrepreneur
```

3. Represent the text

Each person's intro is one document, represented as a bag of words. Counts are converted to tf-idf weights.

```
word_counts <- tokens |> count(wikidata_id, word)

tfidf_long <- word_counts |> bind_tf_idf(word, wikidata_id, n)

keep_words <- word_counts |> count(word) |> filter(n >= 5) |> pull(word)

tfidf_mat <- tfidf_long |>
  filter(word %in% keep_words) |>
  cast_sparse(wikidata_id, word, tf_idf) |>
  as.matrix()

dim(tfidf_mat)  #rows=people, columns=words kept
```

```
## [1] 909 1132
```

Summary of the text:

```
tibble(
  people_with_text = n_distinct(tokens$wikidata_id),
  total_words      = nrow(tokens),
  distinct_words   = n_distinct(tokens$word),
  words_kept_tfidf = length(keep_words)
)
```

```
## # A tibble: 1 x 4
##   people_with_text total_words distinct_words words_kept_tfidf
##           <int>      <int>         <int>         <int>
## 1             909      40388          7589           1132
```

#Length of cleaned intros, women vs men

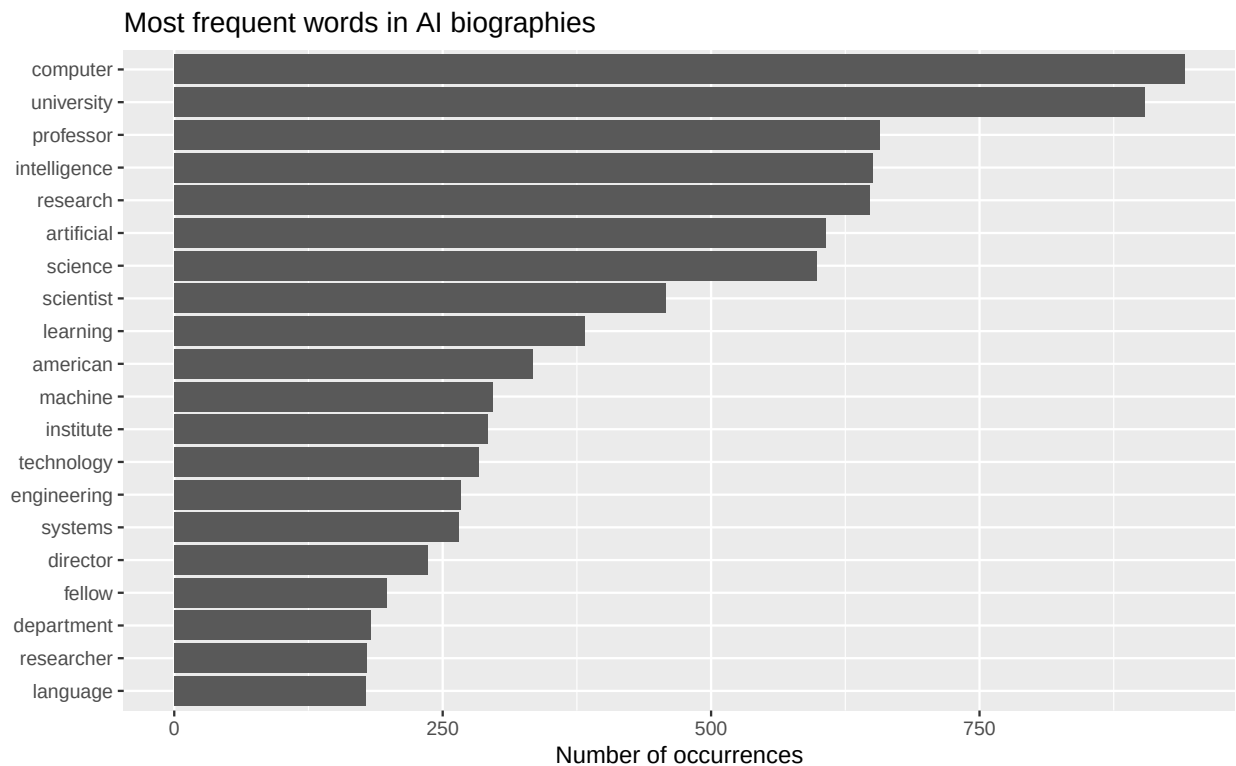
```
tokens |>
  count(wikidata_id, gender) |>
  group_by(gender) |>
  summarise(people = n(),
            median_words = median(n),
            mean_words = round(mean(n), 1),
            .groups = "drop")
```

```
## # A tibble: 2 x 4
##   gender people median_words mean_words
##   <fct>  <int>      <dbl>      <dbl>
## 1 female   218         33        38.8
## 2 male    691         35        46.2
```

4. RQ3: How are women and men described?

Figure 1: most frequent words overall

```
plot_freq <- tokens |>
  count(word, sort = TRUE) |>
  slice_head(n = 20) |>
  ggplot(aes(n, reorder(word, n))) +
  geom_col() +
  labs(title = "Most frequent words in AI biographies",
        x = "Number of occurrences", y = NULL)
plot_freq
```



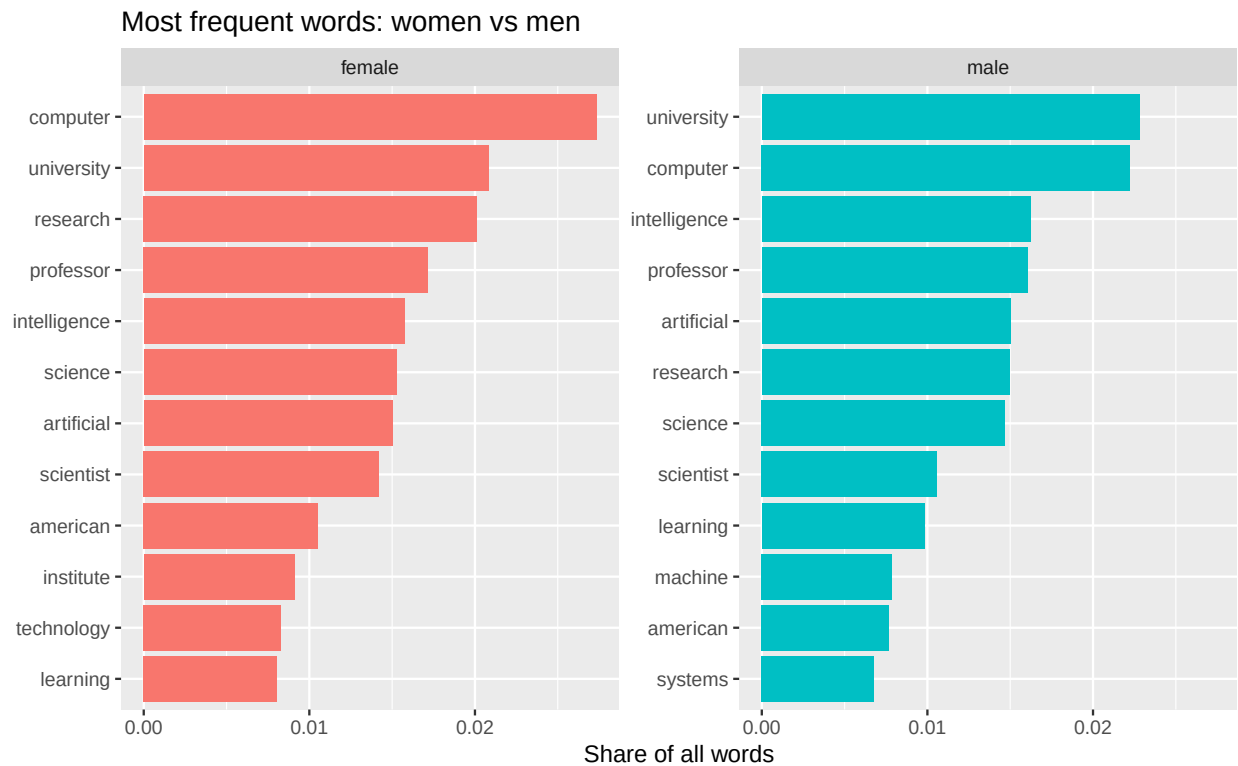
```
ggsave("figures/freq_words_overall.png", plot_freq, width = 6, height = 5, dpi = 300)
```

Figure 2: top words for women vs men

There are far more men than women, so the chart uses each word's share of all words within a gender instead of raw counts.

```
plot_gender <- tokens |>
  count(gender, word) |>
  group_by(gender) |>
  mutate(share = n / sum(n)) |>
  slice_max(share, n = 12, with_ties = FALSE) |>
  ungroup() |>
  ggplot(aes(share, reorder_within(word, share, gender), fill = gender)) +
  geom_col(show.legend = FALSE) +
  scale_y_reordered() +
  facet_wrap(~gender, scales = "free_y") +
  labs(title = "Most frequent words: women vs men",
```

```
x = "Share of all words", y = NULL)
plot_gender
```



```
ggsave("figures/freq_words_gender.png", plot_gender, width = 8, height = 5, dpi = 300)
```

Figure 3: distinctive words by era and gender

Each era and gender group is treated as one document and tf-idf is calculated across the groups. A word only counts if at least `min_people` different people in that group use it.

```
min_people <- 3

group_sizes <- tokens |>
  filter(!is.na(era)) |>
  distinct(wikidata_id, era, gender) |>
  count(era, gender, name = "group_n")
group_sizes
```

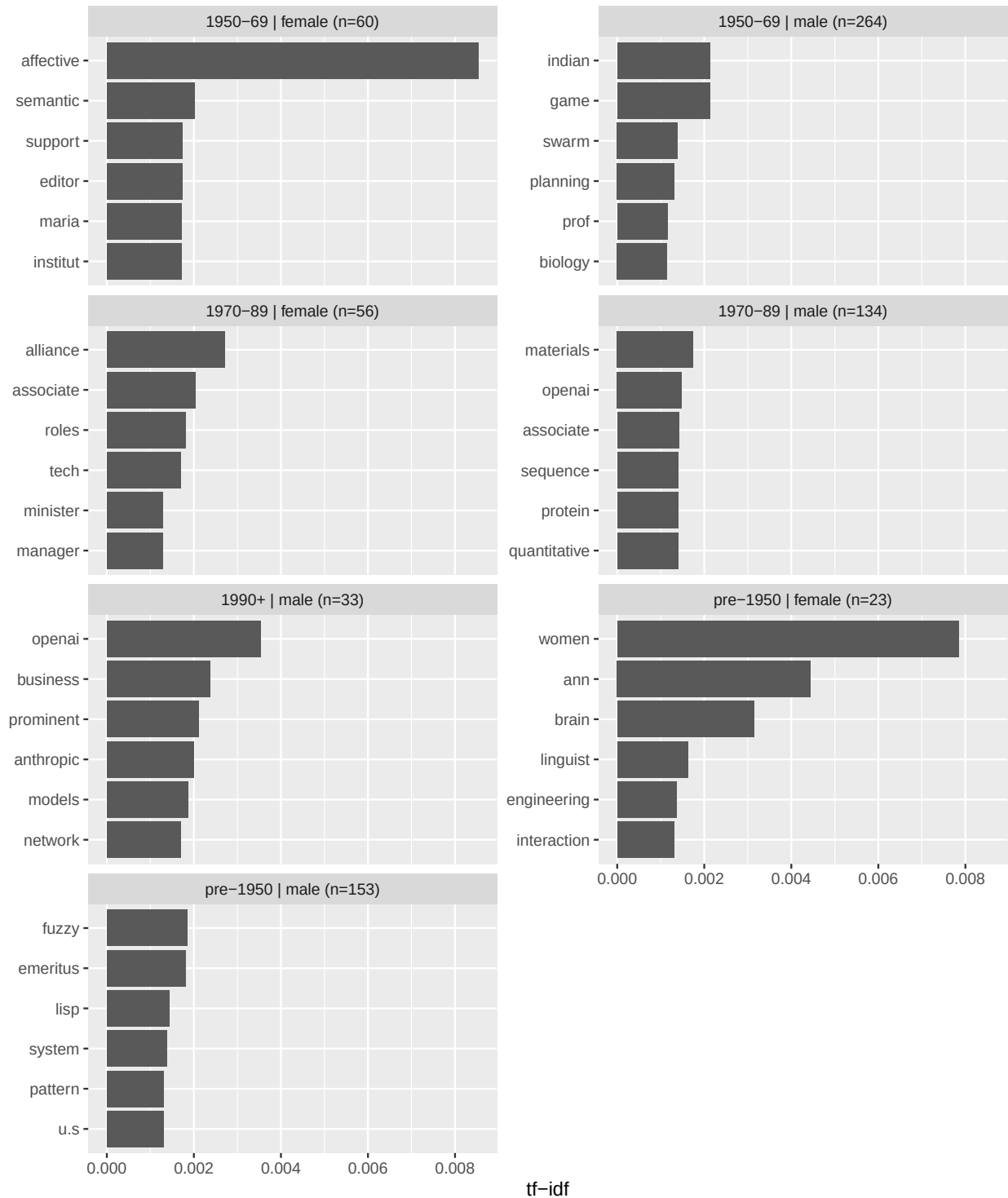
```
## # A tibble: 8 x 3
##   era      gender group_n
##   <fct>   <fct>   <int>
## 1 pre-1950 female     23
## 2 pre-1950 male     153
## 3 1950-69 female     60
## 4 1950-69 male     264
## 5 1970-89 female     56
## 6 1970-89 male     134
```

```
## 7 1990+    female      8
## 8 1990+    male       33
```

```
plot_era <- tokens |>
  filter(!is.na(era)) |>
  mutate(group = paste(era, gender, sep = " | ")) |>
  group_by(group, word) |>
  summarise(n = n(), n_people = n_distinct(wikidata_id), .groups = "drop") |>
  bind_tf_idf(word, group, n) |>
  filter(n_people >= min_people) |>
  group_by(group) |>
  slice_max(tf_idf, n = 6, with_ties = FALSE) |>
  ungroup() |>
  left_join(
    group_sizes |>
      mutate(group = paste(era, gender, sep = " | "),
             group_label = paste0(group, " (n=", group_n, ")")) |>
      select(group, group_label),
    by = "group"
  ) |>
  filter(n_people >= min_people, tf_idf > 0) |>
  ggplot(aes(tf_idf, reorder_within(word, tf_idf, group_label))) +
  geom_col() +
  scale_y_reordered() +
  facet_wrap(~group_label, scales = "free_y", ncol = 2) +
  labs(title = "Most distinctive words by era and gender (tf-idf)",
       subtitle = paste0("Words used by at least ", min_people, " people in the group"),
       x = "tf-idf", y = NULL)
plot_era
```

Most distinctive words by era and gender (tf-idf)

Words used by at least 3 people in the group



```
ggsave("figures/distinctive_words_era_gender.png", plot_era,
       width = 8, height = 10, dpi = 300)
```

Analysis:

RQ3: Is the way Wikipedia describes women in AI different from the way it describes men? Method: Each Wikipedia introduction was split into single words, removing numbers, words of one or two letters, common stop words, a few uninformative words, and each person's own name. We compared word frequencies overall, for women and men separately, and for birth-era and gender groups using tf-idf. All comparisons are descriptive and untested. Across all biographies, the most frequent words are computer, university, professor, intelligence and research (Figure 1), so people in AI are described mainly by their academic positions and affiliations, with american also in the top ten. Women and men share a very similar vocabulary (Figure 2, which uses each word's share within a gender because there are far more men): university, research, professor and intelligence are in both top tens. The differences are small and untested: research, scientist, institute and technology take a slightly larger share of women's introductions, while machine and systems appear among men's top words but not women's. By era (Figure 3, words used by at least three people in a group), alliance, minister, manager and roles distinguish women born 1970-89, and women is the most distinctive word for those born before 1950. Men born before 1950 are distinguished by classic AI terms (fuzzy, lisp, pattern), and men born 1990 or later by industry terms (openai, anthropic, business, models). Women born 1990 or later do not appear in Figure 3, because no distinguishing word met the threshold.

5. RQ4: Which AI subfields do women and men appear in, and how does the distribution differ?

5.1: Data representation and distance: Each person is a row of tf-idf weights over the kept words. Each row is scaled to length 1 so that longer intros do not look different just because of their length. With unit-length rows, Euclidean distance ranks pairs of people the same way as cosine distance.

```
row_norm <- sqrt(rowSums(tfidf_mat^2))
m_norm   <- tfidf_mat[row_norm > 0, , drop = FALSE] / row_norm[row_norm > 0]
dim(m_norm)
```

```
## [1] 909 1132
```

5.2: Choosing the number of clusters (k)

Elbow plot and average silhouette width for k = 2 to 10.

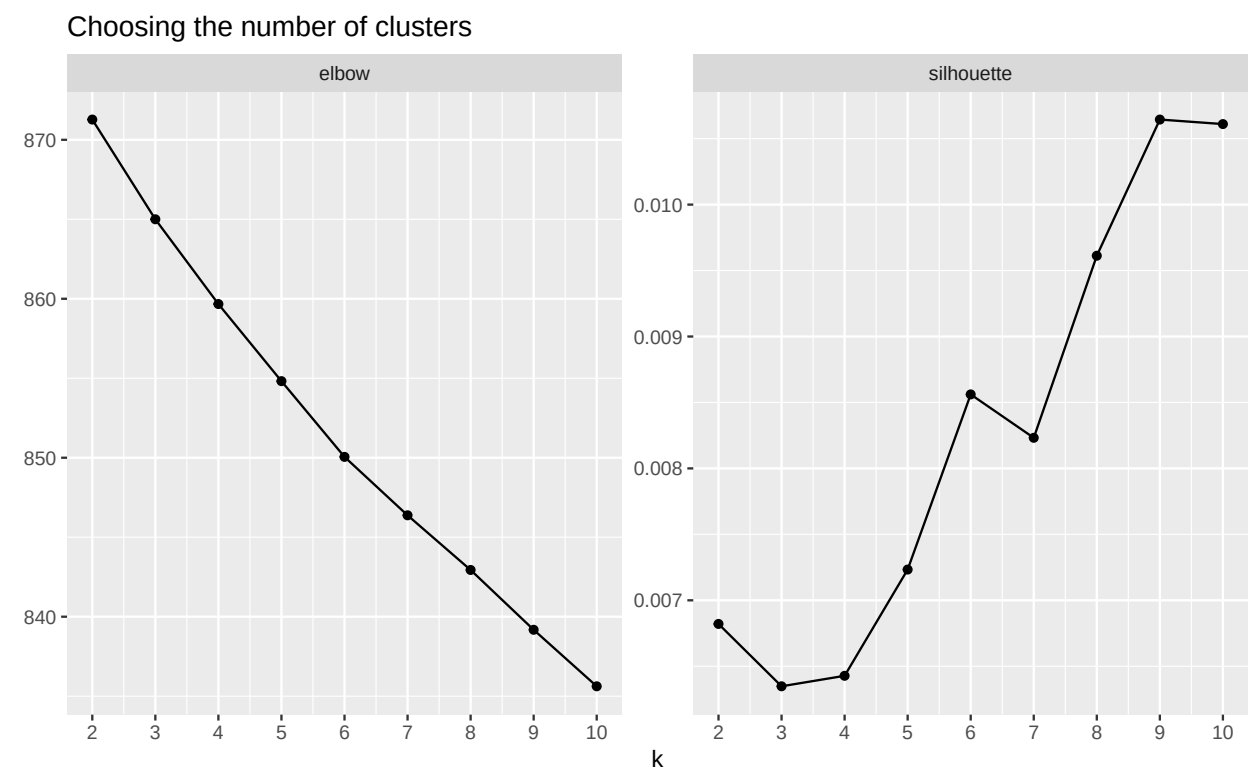
```
set.seed(1)
ks <- 2:10
fits <- map(ks, \(k) kmeans(m_norm, centers = k, nstart = 10, iter.max = 50))
d <- dist(m_norm)

k_check <- tibble(
  k = ks,
  elbow = map_dbl(fits, "tot.withinss"),
  silhouette = map_dbl(fits, \(f) mean(silhouette(f$cluster, d)[, 3]))
)
k_check
```

```
## # A tibble: 9 x 3
##       k elbow silhouette
##   <int> <dbl>     <dbl>
## 1     2  871.     0.00682
## 2     3  865.     0.00635
## 3     4  860.     0.00643
## 4     5  855.     0.00723
## 5     6  850.     0.00856
```

```
## 6      7 846.    0.00823
## 7      8 843.    0.00961
## 8      9 839.    0.0106
## 9     10 836.    0.0106
```

```
p_k <- k_check |>
  pivot_longer(c(elbow, silhouette), names_to = "measure") |>
  ggplot(aes(k, value)) +
  geom_line() + geom_point() +
  facet_wrap(~measure, scales = "free_y") +
  scale_x_continuous(breaks = ks) +
  labs(title = "Choosing the number of clusters", y = NULL)
p_k
```



```
ggsave("figures/choose_k.png", p_k, width = 8, height = 4, dpi = 300)
```

5.3: Run the clustering

```
K <- 5 # chosen for interpretability
set.seed(1)
km <- kmeans(m_norm, centers = K, nstart = 25, iter.max = 100)
table(km$cluster)
```

```
##
## 1 2 3 4 5
## 141 68 117 107 476
```

5.4: What is each cluster about?

```
cluster_words <- map_dfr(seq_len(K), \(i) {  
  ctr <- km$centers[i, ]  
  tibble(cluster = i,  
    size = sum(km$cluster == i),  
    top_words = paste(names(sort(ctr, decreasing = TRUE))[1:8], collapse = ", "))  
})  
cluster_words
```

```
## # A tibble: 5 x 3  
##   cluster size top_words  
##   <int> <int> <chr>  
## 1      1  141 learning, machine, statistics, neural, computer, science, scien~  
## 2      2   68 language, linguistics, natural, computational, processing, ling~  
## 3      3  117 founder, company, chinese, founded, entrepreneur, ceo, chief, o~  
## 4      4  107 engineering, electrical, ieee, fellow, systems, engineers, elec~  
## 5      5  476 artificial, intelligence, computer, science, university, resear~
```

```
ctr_norm <- km$centers / sqrt(rowSums(km$centers^2))  
cos_sims <- m_norm %*% t(ctr_norm)  
  
examples <- map_dfr(seq_len(K), \(i) {  
  members <- which(km$cluster == i)  
  top <- members[order(cos_sims[members, i], decreasing = TRUE)][seq_len(min(3, length(members)))]  
  tibble(cluster = i, wikidata_id = rownames(m_norm)[top])  
}) |>  
  left_join(select(people, wikidata_id, name), by = "wikidata_id") |>  
  group_by(cluster) |>  
  summarise(closest_people = paste(name, collapse = "; "))  
examples
```

```
## # A tibble: 5 x 2  
##   cluster closest_people  
##   <int> <chr>  
## 1      1 Barbara Hammer (computer scientist); Frank Hutter; John Shawe-Taylor  
## 2      2 Vera Demberg; Emily M. Bender; Lori Levin  
## 3      3 Robin Li; Yan Junjie; Amnon Shashua  
## 4      4 Fakhreddine Karay; Hao Ying; Pramod Varshney  
## 5      5 Nello Cristianini; Takayuki Ito; Rolf Pfeifer
```

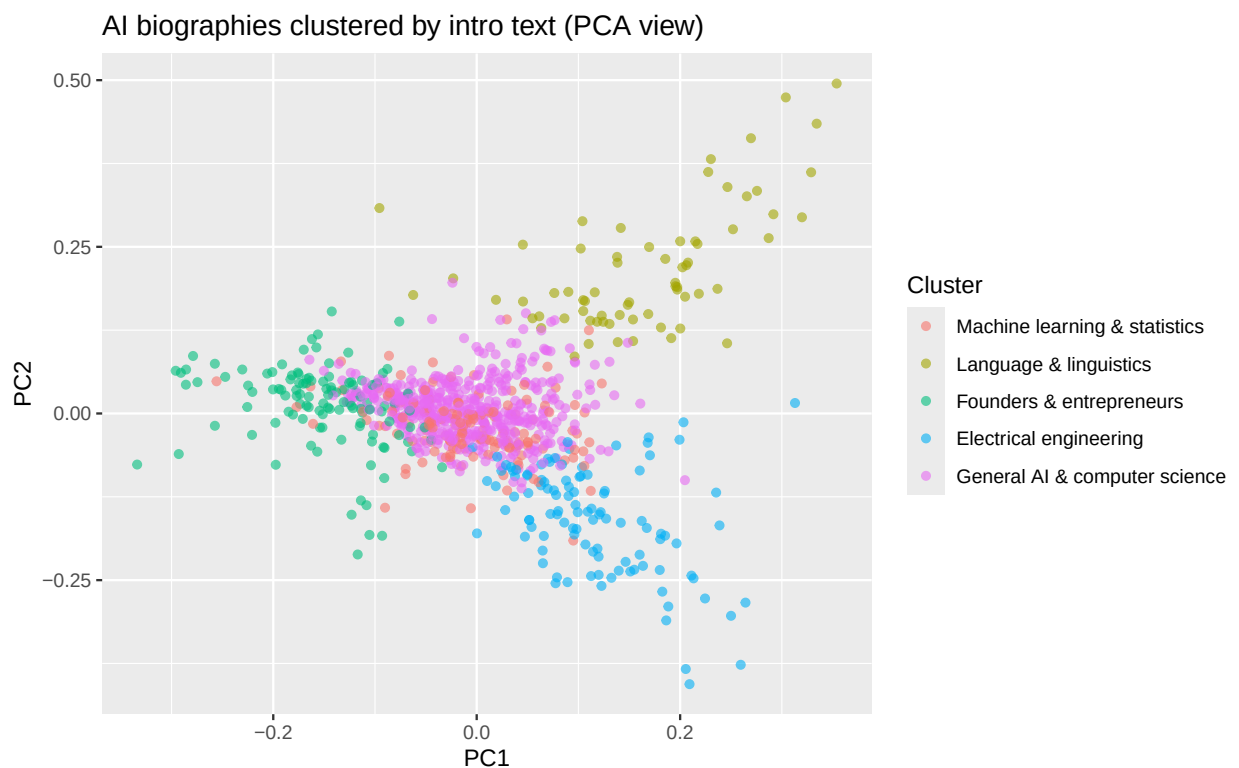
Cluster names, in order 1 to 5:

```
cluster_names <- c("Machine learning & statistics",  
  "Language & linguistics",  
  "Founders & entrepreneurs",  
  "Electrical engineering",  
  "General AI & computer science")
```

5.5: Visualise the clusters: PCA splits the many word dimensions into 2 so the clusters can be drawn. It shows only part of the structure.

```
clusters <- tibble(wikidata_id = rownames(m_norm),
                  cluster = km$cluster) |>
  left_join(select(people, wikidata_id, name, gender, era), by = "wikidata_id") |>
  mutate(cluster_label = factor(cluster, levels = seq_len(K), labels = cluster_names))

pc <- prcomp(m_norm, rank. = 2)
p_clusters <- clusters |>
  mutate(PC1 = pc$x[, 1], PC2 = pc$x[, 2]) |>
  ggplot(aes(PC1, PC2, colour = cluster_label)) +
  geom_point(alpha = 0.6) +
  labs(title = "AI biographies clustered by intro text (PCA view)",
       colour = "Cluster")
p_clusters
```



```
ggsave("figures/clusters_pca.png", p_clusters, width = 7, height = 5, dpi = 300)
```

Check-up:

```
nrow(m_norm); nrow(people_text)           # people reduced before clustering
```

```
## [1] 909
```

```
## [1] 910
```

```
nrow(clusters) == nrow(m_norm)
```

```
## [1] TRUE
```

```
anyDuplicated(people$wikidata_id)
```

```
## [1] 0
```

```
sum(is.na(clusters$era)) # people with no recorded era
```

```
## [1] 178
```

5.6: Women and men across clusters: Chi-square test of independence on the cluster-by-gender table. We have to check the expected counts first.

```
tab <- table(clusters$cluster_label, clusters$gender)
tab
```

```
##
##               female male
## Machine learning & statistics    24  117
## Language & linguistics           31   37
## Founders & entrepreneurs          23   94
## Electrical engineering            22   85
## General AI & computer science    118  358
```

```
round(chisq.test(tab)$expected, 1) # ideally all values are 5 or more
```

```
##
##               female male
## Machine learning & statistics   33.8 107.2
## Language & linguistics          16.3  51.7
## Founders & entrepreneurs         28.1  88.9
## Electrical engineering           25.7  81.3
## General AI & computer science   114.2 361.8
```

```
chisq.test(tab)
```

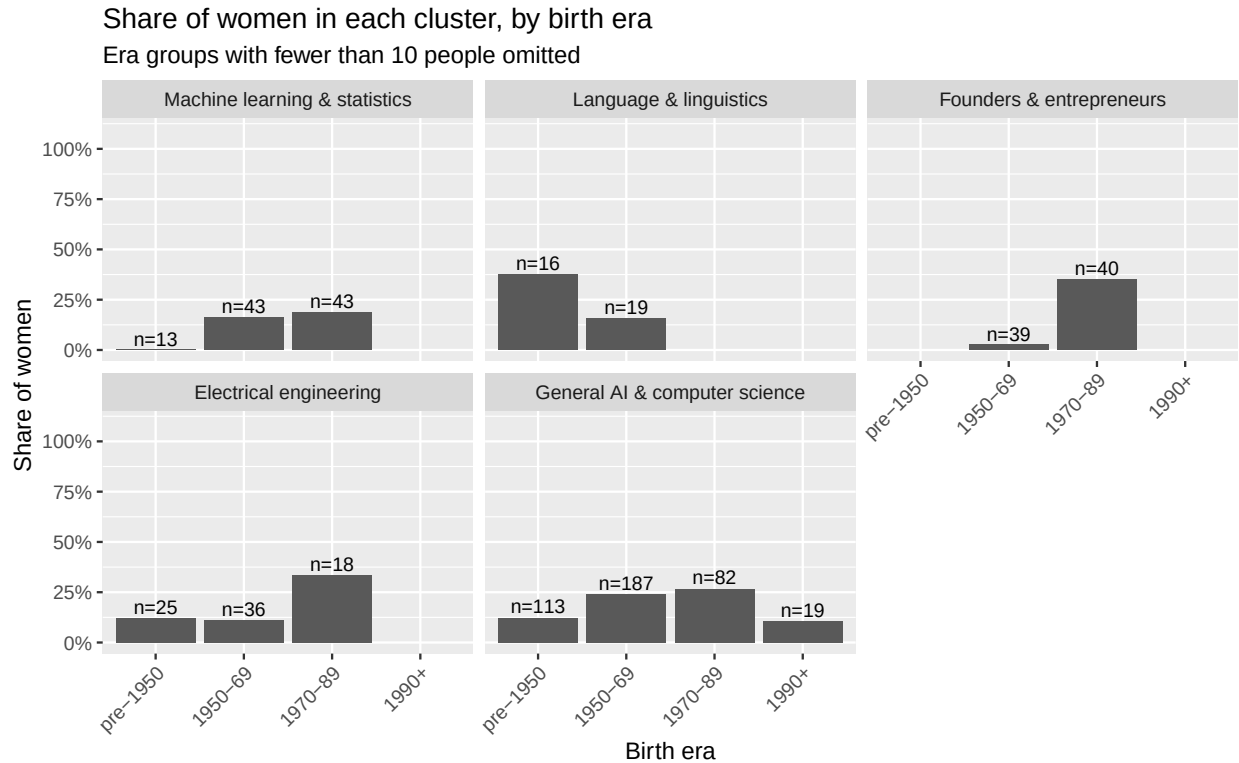
```
##
## Pearson's Chi-squared test
##
## data:  tab
## X-squared = 23.217, df = 4, p-value = 0.0001146
```

Women's share in each cluster, by birth era. Groups with fewer than 10 people are abolished because small groups give unreliable shares.

```
share_cluster_era <- clusters |>
  filter(!is.na(era)) |>
  group_by(cluster_label, era) |>
  summarise(n = n(), share_women = mean(gender == "female"), .groups = "drop") |>
  filter(n >= 10)
share_cluster_era
```

```
## # A tibble: 14 x 4
##   cluster_label      era      n share_women
##   <fct>          <fct> <int>      <dbl>
## 1 Machine learning & statistics pre-1950    13      0
## 2 Machine learning & statistics 1950-69    43    0.163
## 3 Machine learning & statistics 1970-89    43    0.186
## 4 Language & linguistics      pre-1950    16    0.375
## 5 Language & linguistics      1950-69    19    0.158
## 6 Founders & entrepreneurs     1950-69    39    0.0256
## 7 Founders & entrepreneurs     1970-89    40    0.35
## 8 Electrical engineering      pre-1950    25    0.12
## 9 Electrical engineering      1950-69    36    0.111
## 10 Electrical engineering     1970-89    18    0.333
## 11 General AI & computer science pre-1950   113    0.124
## 12 General AI & computer science 1950-69   187    0.241
## 13 General AI & computer science 1970-89    82    0.268
## 14 General AI & computer science 1990+     19    0.105
```

```
p_share <- ggplot(share_cluster_era, aes(era, share_women)) +
  geom_col() +
  geom_text(aes(label = paste0("n=", n)), vjust = -0.3, size = 3) +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1.1),
    breaks = c(0, 0.25, 0.5, 0.75, 1)) +
  facet_wrap(~cluster_label) +
  labs(title = "Share of women in each cluster, by birth era",
    subtitle = "Era groups with fewer than 10 people omitted",
    x = "Birth era", y = "Share of women") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
p_share
```



```
ggsave("figures/share_women_cluster_era.png", p_share,
       width = 8, height = 6, dpi = 300)

saveRDS(clusters, "data/clusters.rds")
```

Analysis: RQ4: Which AI subfields do women and men appear in, and how does the distribution differ? Method: Each cleaned introduction became a tf-idf vector over words used by at least five people, scaled to length 1 so introduction length does not matter (making Euclidean distance equivalent to cosine distance). We clustered the 909 people with usable text using k-means (k chosen from 2-10 by the elbow and silhouette width, fixed seed), described each cluster by its top words and closest people, and tested cluster-by-gender association with a chi-square test, then examined women's share by birth era. Choosing the number of clusters: The elbow plot has no bend, and silhouette widths are low for every k (all < 0.011), so the clusters overlap. The highest values were at k = 9 and 10, but the differences are tiny. We chose k = 5 because the clusters were interpretable and large enough to compare by gender and era. The clusters: Machine learning and statistics (n = 141): learning, machine, statistics, neural (e.g. Barbara Hammer). Language and linguistics (n = 68): language, linguistics, natural, computational (e.g. Emily M. Bender). Founders and entrepreneurs (n = 117): founder, company, entrepreneur, ceo, grouping people by business role (e.g. Robin Li). Electrical engineering (n = 107): engineering, electrical, ieee, systems (e.g. Pramod Varshney). General AI and computer science (n = 476), the largest, looks like a catch-all for academics described in general AI terms. The PCA plot shows heavy overlap in the centre, matching the low silhouette values. Women and men across clusters: Women are 24.0% of the 909 people but 45.6% of Language and linguistics (31 of 68) and 17.0% of Machine learning and statistics (24 of 141). The association is statistically significant, $\chi^2(4, N = 909) = 23.2, p < 0.001$, and all expected counts were at least 5. Women's share by birth era: Groups with fewer than 10 people and the 178 people with no birth era are omitted. Women's share is higher for people born 1970-89 than earlier in three clusters: Founders and entrepreneurs (about 3% to 35%), Electrical engineering (11-12% to 33%, from only 18 people), and General AI and computer science (12% to 24-27%). Machine learning and statistics stays low throughout (0-18%), and Language and linguistics has too few people in recent eras to show change. These patterns are descriptive and based on small groups.